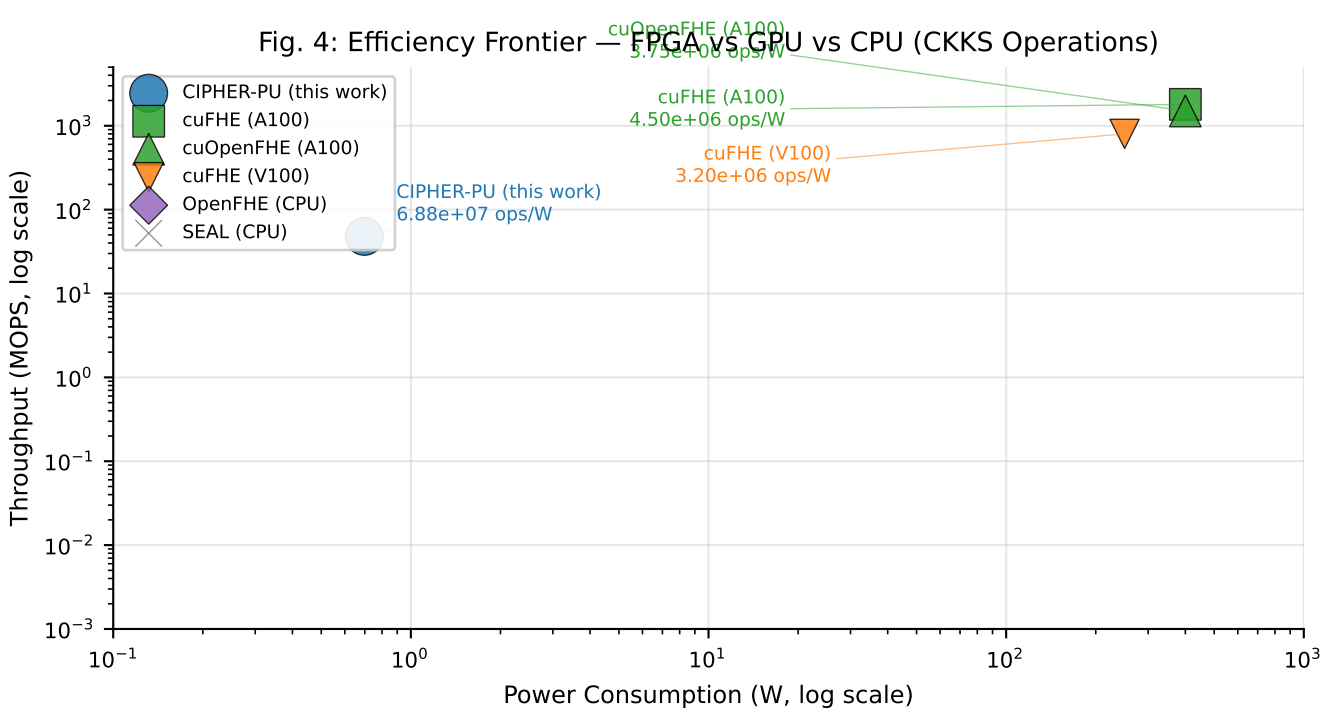
Fig. 4: Efficiency Frontier — FPGA vs GPU vs CPU (CKKS Operations)
CIPHER-PU (this work)
cuFHE (A100)
cuOpenFHE (A100)
cuFHE (V100)
OpenFHE (CPU)
SEAL (CPU)
cuOpenFHE (A100)
3.75e+08 ops/W
cuFHE (A100)
4.50e+06 ops/W
cuFHE (V100)
3.20e+06 ops/W
CIPHER-PU (this work)
6.88e+07 ops/W
Throughput (MOPS, log scale)
Power Consumption (W, log scale)
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}